# Johnny Dunn

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## SUMMARY

Full-stack engineer in applied AI/ML (NLP, computer vision): LLM agents and tool use, RAG, evals, fine-tuning, model serving, deep learning, GPU training, quantization, edge inference, IVRs. Seven years end to end, the last three and a half as an independent contractor.

## PROFESSIONAL EXPERIENCE

### Machine Learning Engineer
Mar 2023 – Present

*Independent Consulting (Contract) — Los Angeles, CA*

- Tech lead on Aye Eye Global's edge AI vision product: Go daemon on a Rockchip RK3588 board, multi-camera streams, WebRTC, NPU inference (PyTorch to ONNX to INT8), quantized LLM voice interface, React / TypeScript dashboard; live with 50+ users in UK retail.
- Improved AI text detection accuracy from 55% to 71% for Grapple Research by fine-tuning GPT-3.5; built the training scrapers, deployed with Docker and Supabase on AWS, and cut inference costs by 64% with document summarization and caching.
- Built a semantic similarity matching tool for ImmenseX (stealth SaaS) on hybrid vector / graph RAG (Weaviate, Neo4j), served open-weights models with vLLM on GCP / Linux, and reduced the client's prompt-writing time by 60%.
- Built Linkby's Python ML pipeline on Databricks / Spark, orchestrated with Airflow: automated feature engineering and model training from Snowflake ad data, running RF, SVM, Naive Bayes and LR in parallel under an A/B testing framework.

### Senior Blockchain Engineer
Nov 2021 – Feb 2023

*Tilting Point — New York, NY*

- Acted as blockchain tech lead: created ERC-20 / 721 upgradable proxy smart contracts, built containerized REST / gRPC marketplace APIs with SSO and an airdrop tool, and rescued $1M+ in MATIC by reverse engineering a Gnosis Safe.
- Architected Go microservices streaming on-chain and off-chain events through Kafka for Unity SDKs in 4+ games; 1000+ TPS analytics.

### Software Engineer 2
Jun 2020 – Apr 2021

*eBay Inc. — New York, NY*

- Built REST APIs in Scala / Spring Boot streaming new ads in real time through Kafka from selected item-option clicks (under 1s latency).
- Owned end-to-end MLOps infrastructure (CI/CD with Kubernetes, Jenkins, MLflow) so teams could deploy models to sandboxes exposing data-science-driven statistics; accelerated deep learning model iteration by 50% and built scrapers verifying data integrity.

### Machine Learning Engineer
Jan 2018 – May 2020

*Independent Consulting (Contract) — New York, NY*

- Reached 77% sentiment analysis accuracy with a random forest trained on 10K texts for HereAfterLegacy.ai; built the MVP with scrapers for TV scripts, recreated its characters as digital personas, and deployed it on Google Cloud Platform.
- Trained neural style transfer models behind a REST API exhibit applying film-inspired filters to selfies for Edelman.
- Delivered user research, animated Figma wireframes and a working MVP demo for Notebrush.

### Cognitive Implementation Engineer
Jun 2017 – Dec 2017

*Amelia (formerly IPsoft Inc.) — New York, NY*

- Built AI assistant enterprise modules (PII redaction for HIPAA, SOC 2 compliance, Agile practices) for Fortune 500 medical suppliers.

## OPEN-SOURCE PROJECTS

**AgentOS** – TypeScript agent runtime (RAG, voice); 680+ stars; 85.6% LongMemEval-S; powers wilds.ai. **github.com/framerslab/agentos**
**Paracosm** – AI agent swarm simulation engine as a structured world model, built on AgentOS. **github.com/framerslab/paracosm**
**tenets** – MCP server for coding agents that aggregates prompt context with neural ranking, fully offline. **github.com/jddunn/tenets**
**Restless** – Hierarchical attention network for anti-malware detection; 93% accuracy on the ClaMP dataset. github.com/jddunn/restless

## EDUCATION

**Bachelor of Fine Arts in Game Design — Parsons School of Design, 2017**

## AWARDS

5th Place, AT&T National VR and AR Challenge, with THiNKIMMERSIVE team (npe100.com / mobile app), 2017.

## SKILLS

**Languages:** Python, TypeScript, JavaScript, Go, Scala, SQL, Bash
**AI / ML:** LLM agents and tool use (MCP, DSPy), RAG, evals, fine-tuning, vLLM serving, PyTorch, scikit-learn, XGBoost, OpenAI, Anthropic
**Web:** React, Next.js, React Native, Node.js, FastAPI, GraphQL, Stripe
**Data:** PostgreSQL (pgvector), Redis, MongoDB, Elasticsearch, Kafka
**Cloud & DevOps:** AWS, GCP, Terraform, Docker, Kubernetes, GitHub Actions, CI/CD